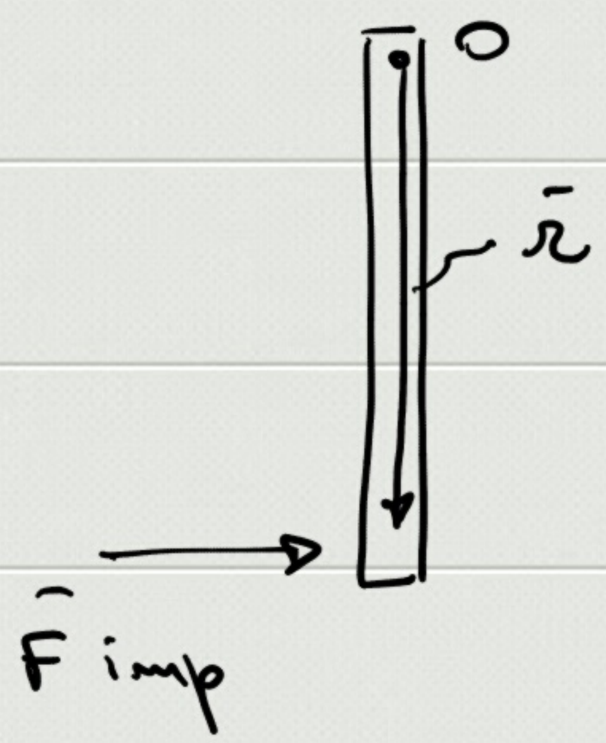
$$\bar{F} = \frac{d\bar{p}}{dt}$$

$$\bar{J} = \int \bar{F}\,dt \overset{!}{=} \int d\bar{p} = \Delta\bar{p}$$

$$\bar{M}_o^E = \frac{d\bar{L}_o}{dt}$$

$$\underbrace{\int_{t_0}^{t} \bar{M}_o^E\,dt = \int_{\bar{L}_o(t_0)}^{\bar{L}_o(t)} d\bar{L}_o = \bar{L}_o(t) - \bar{L}_o(t_0) = \Delta\bar{L}}$$

Teorema del impulso del momento
(impulso angolare)



$$\bar{M}_o = \bar{r} \times \bar{F}$$

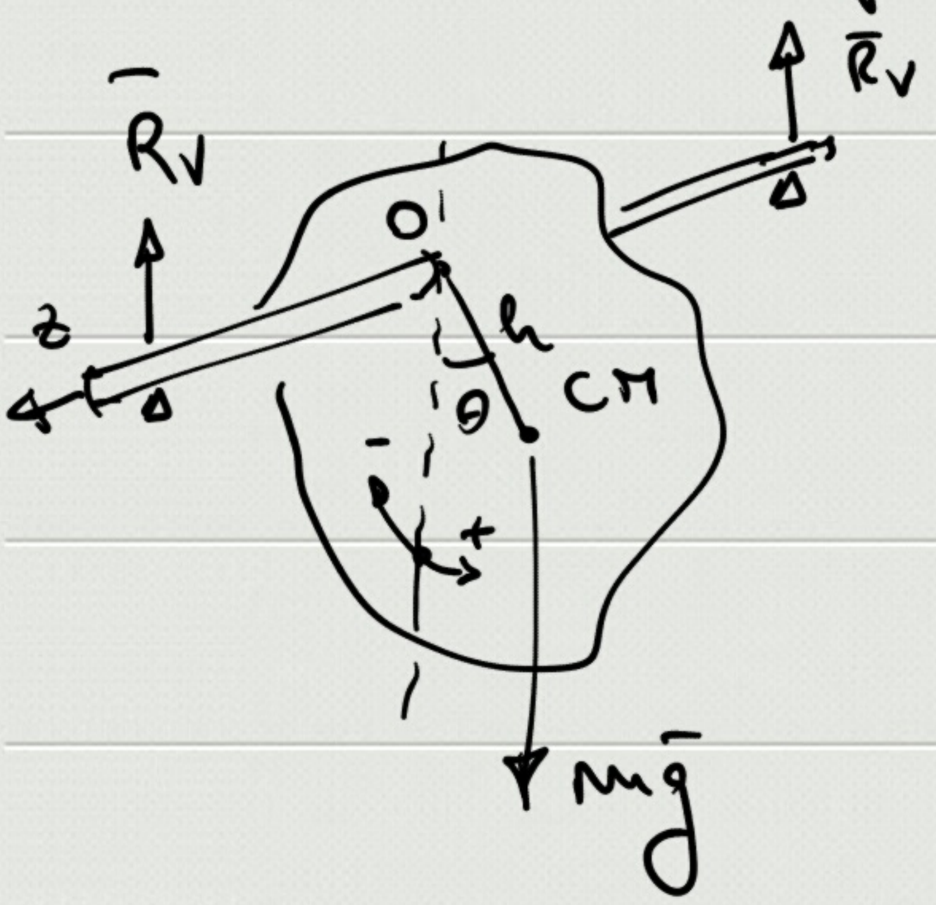
$$\bar{r} \simeq \text{cost}$$

$$\underline{\underline{\int \bar{M}_o\,dt}} = \int \bar{r} \times \bar{F}\,dt = \bar{r} \times \int \bar{F}\,dt = \underline{\underline{\bar{r} \times \bar{J}}}$$

Impulso del momento = momento dell'impulso

Teorema del momento dell'impulso

Pendolo composto



$$\bar{M}^{E}_{O} = I_z \bar{\alpha}$$

$$\bar{M}_{O,peso} = \bar{h} \times m\bar{g}$$

$$\overset{!}{=} h\, mg \sin\theta \,(-\bar{u}_z)$$

$$-mgh \sin\theta = I_z \alpha$$

piccole oscillazioni

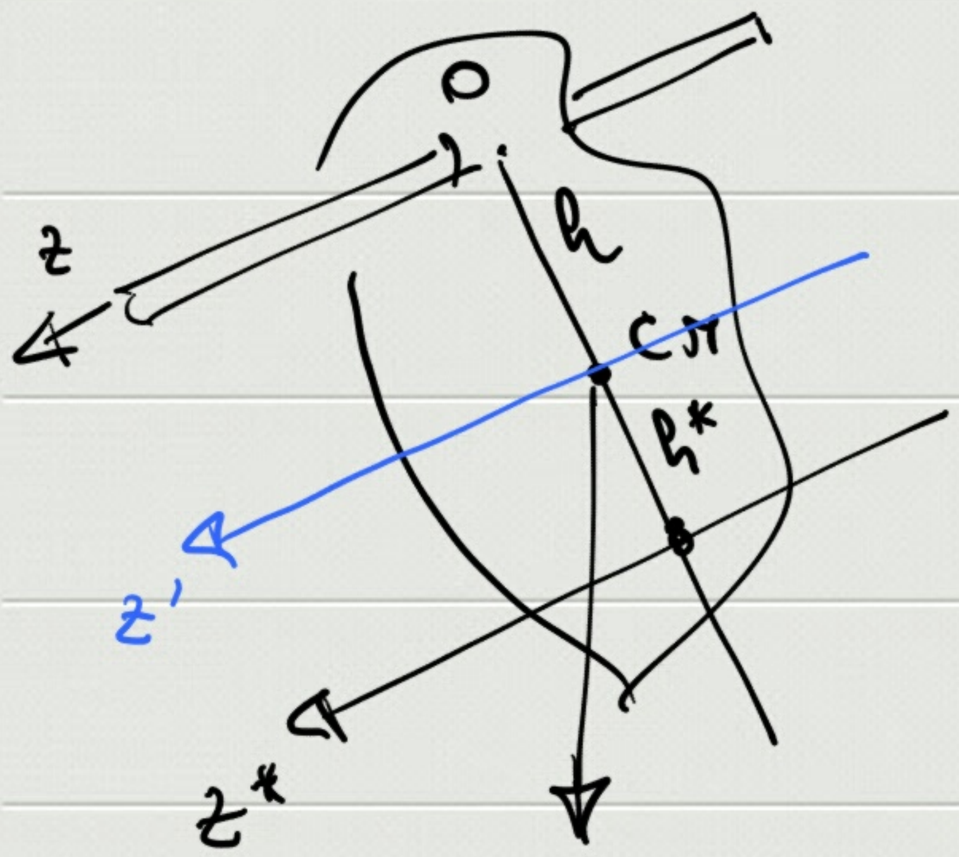
$$\Rightarrow \quad -mgh\,\theta = I_z \frac{d^2\theta}{dt^2}$$

$$\Rightarrow \quad \frac{d^2\theta}{dt^2} + \frac{mgh}{I_z}\theta = 0 \qquad \Omega = \sqrt{\frac{mgh}{I_z}}$$

$$\Rightarrow \quad \boxed{\theta(t) = \theta_0 \sin(\Omega t + \phi)}$$

$$T = \frac{2\pi}{\Omega} = 2\pi\sqrt{\frac{I_z}{mgh}} \qquad T = 2\pi\sqrt{\frac{l}{g}}$$

$$\boxed{l = \frac{I_z}{mh}}$$ lunghezza ridotta



$m\bar{g}$

H-S

$$l = \frac{I_z}{mh} \overset{!}{=} \frac{I_{z'} + mh^2}{mh} =$$

$$= \frac{I_{z'}}{mh} + h = h^* + h$$

$$\Rightarrow \boxed{I_{z'} = mhh^*}$$

$$\Rightarrow l^* = \frac{I_{z^*}}{mh^*}$$

$$l^* = \frac{I_{z^*}}{mh^*} = \frac{I_{z'} + mh^{*2}}{mh^*} = \frac{I_{z'}}{mh^*} + h^* = h + h^*$$

z e z^* → essi reciproci